

Power and Stokes phase sweep — run report

RUN

010250_stokes-phase-sweep_stokes-phase-sweep

Status: completed | Started: 2026-09-24T00:02:50.630288+00:00

PURPOSE

Measure synchronized optical power and Stokes parameters while sweeping an external phase actuator. Repeated OUT1 sine cycles; IN1 is the measured voltage reference, with acquisition timing retained.

REQUIRED OPTICAL SETUP

PAX in path A / E4, observing the combined field. OUT1 drives phi1 and is physically T-ed into IN1 as the voltage reference (not a photodiode). OUT2 stays at zero.

RECORDED BENCH CONTEXT (RECIPE)

In front of NPBS 1, C port

Red Pitaya (0-1 V command) -> Thorlabs MDT690 -> piezo

OPERATOR SETUP NOTES

V_pi expectation only; verify channel and measured driver gain before calibration.

DATA

349 recorded samples; acquisition: 349

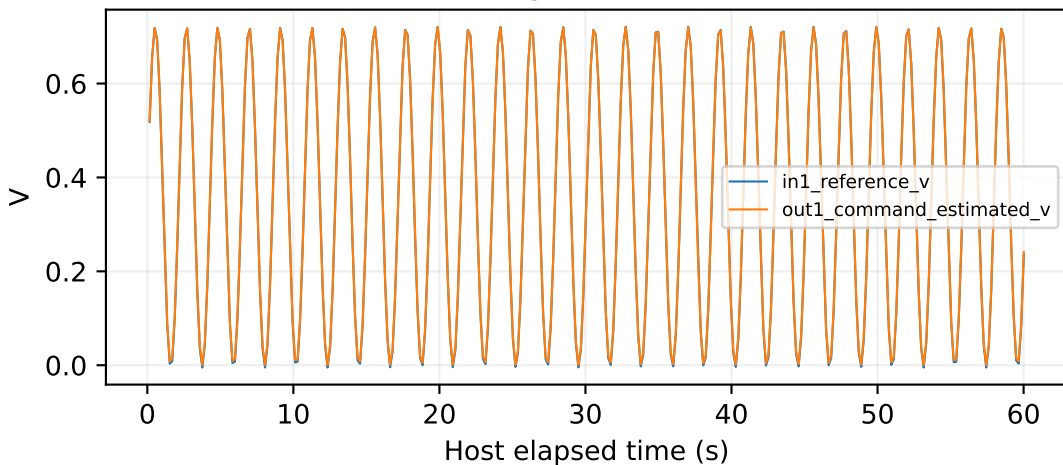
Source: data.csv; full acquisition settings: recipe.json; status: run.json

RAW DOP

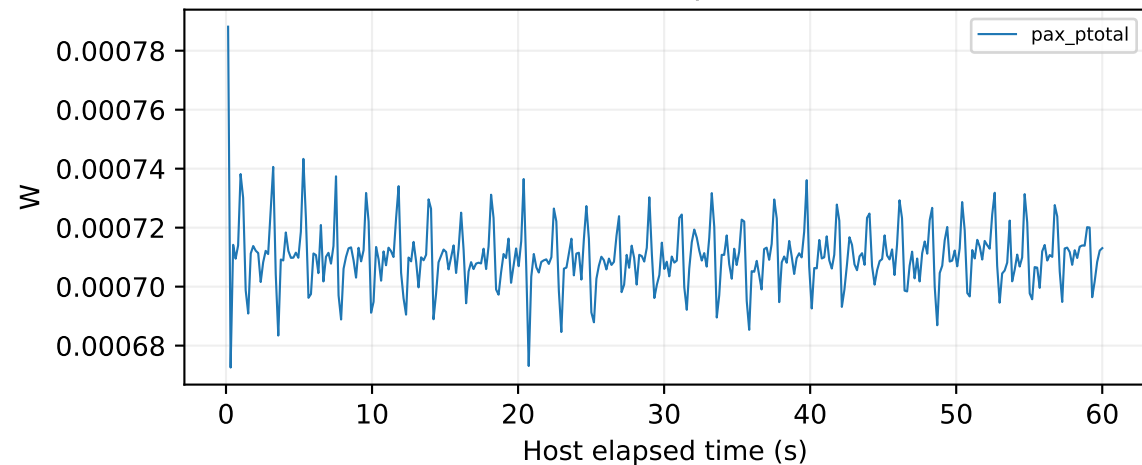
Mean 0.9183; range 0.4377 to 0.9764. 0 samples outside 0–1. Telemetry is not a new calibration.

Power and Stokes versus phase-actuator drive

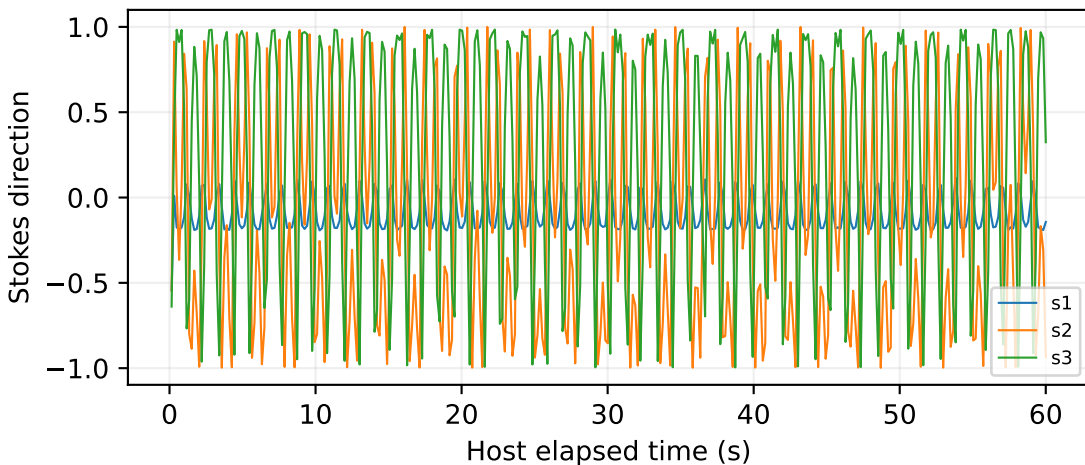
Voltage reference



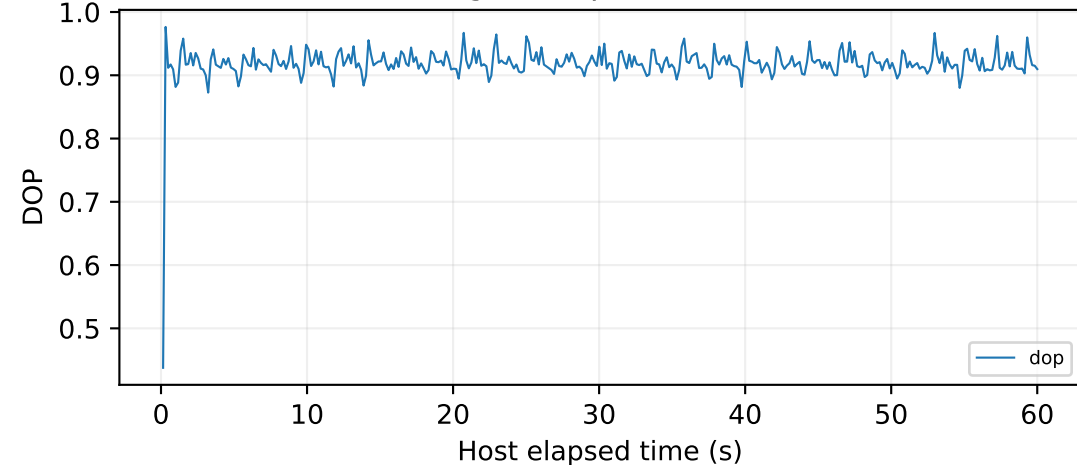
PAX total power



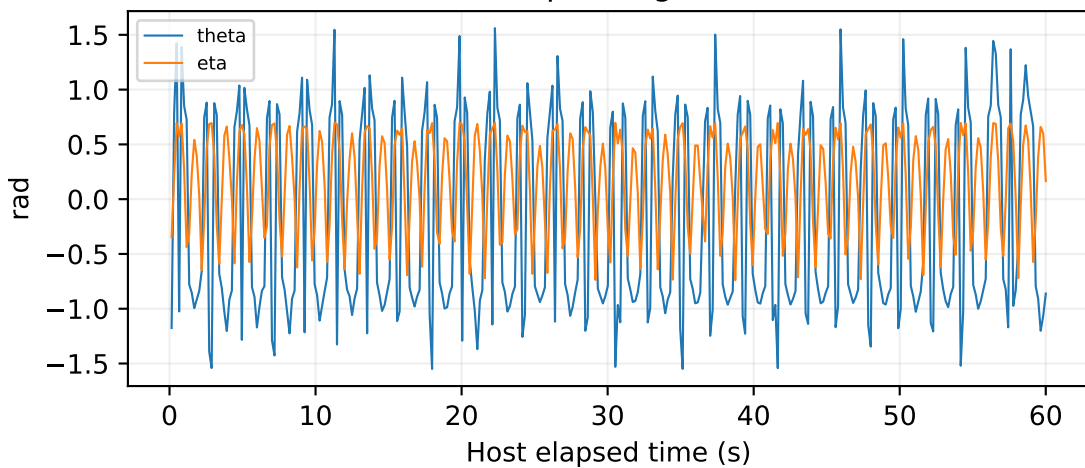
Polarization direction (unit norm)



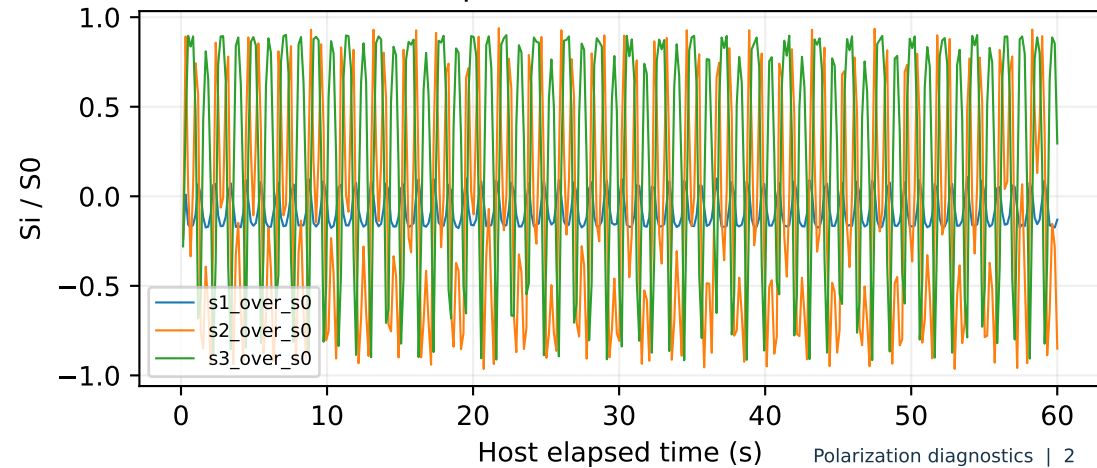
Degree of polarization



Ellipse angles



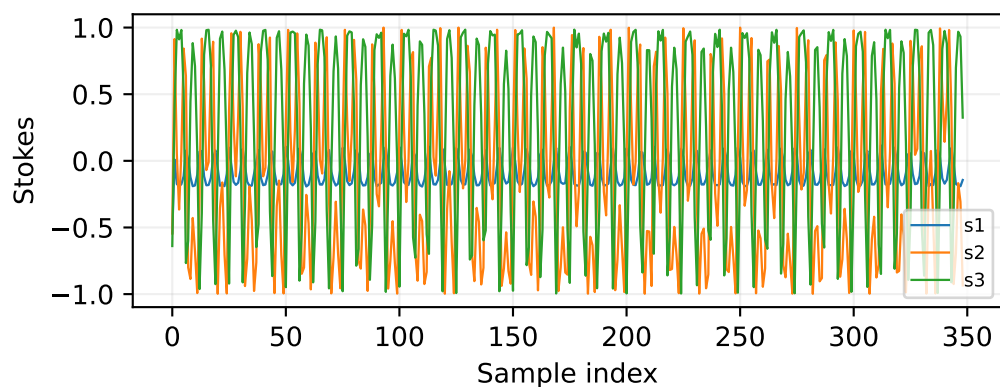
Total-power-normalized Stokes



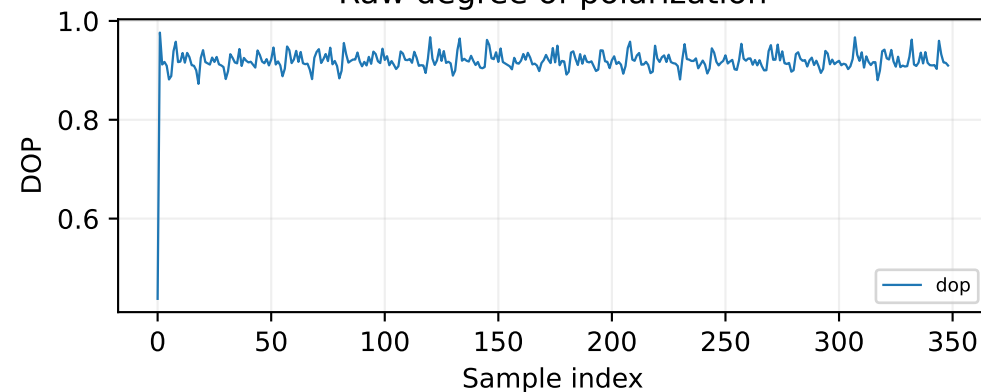
Recorded telemetry

Acquisition order preserves stage changes; no timing alignment or transfer gain is inferred.

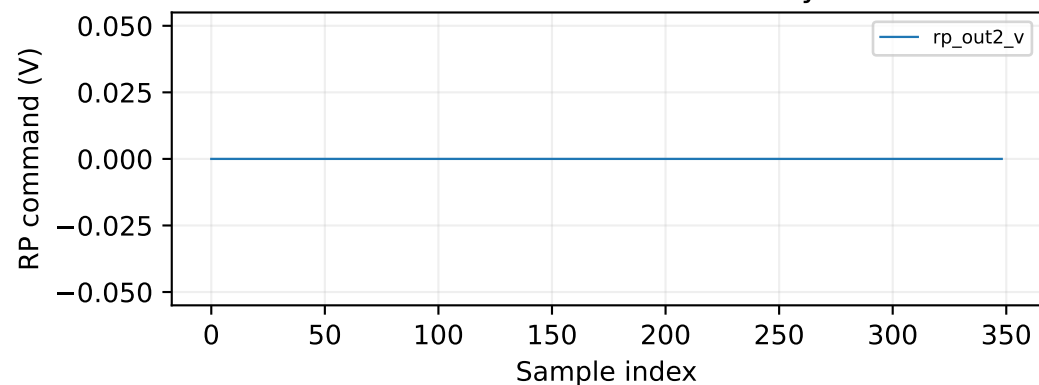
Normalized Stokes



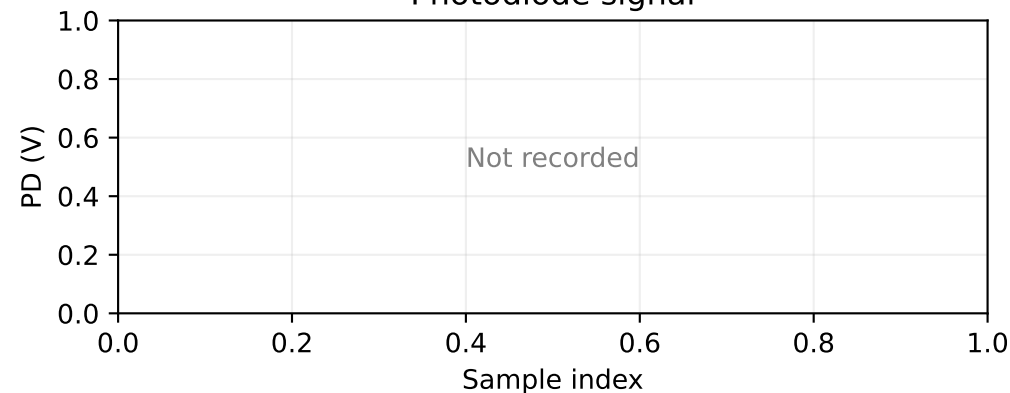
Raw degree of polarization



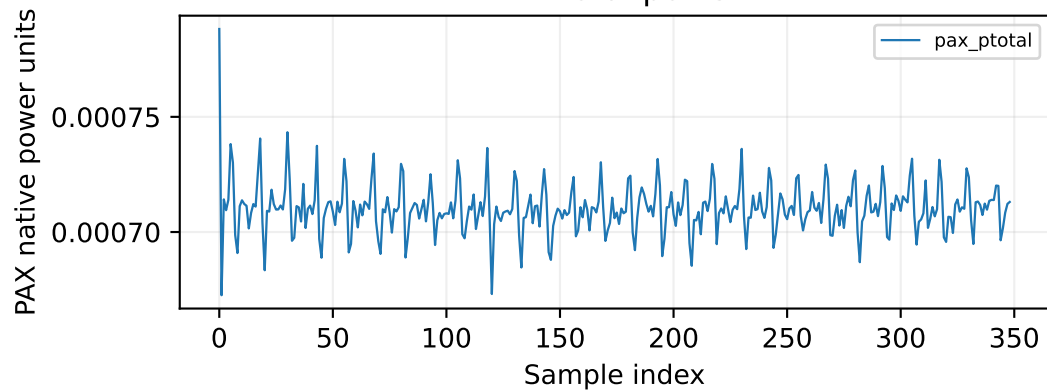
Actuator command history



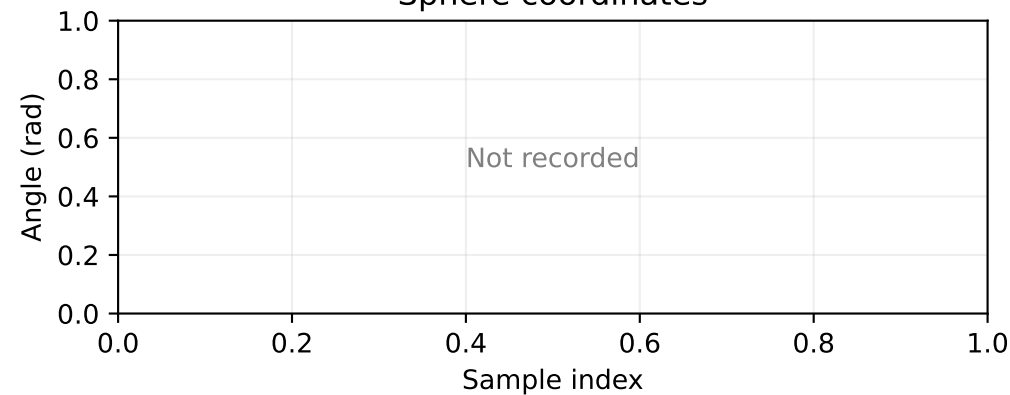
Photodiode signal



PAX total power



Sphere coordinates



Active parameters

RP commands: volts, bounded to 0–1. V_lambda: terminal volts for 2π. Gains: terminal V / RP V.

Parameter	Recorded value
label	"stokes-phase-sweep"
duration_s	60.0
report	"pdf"
pax_rotation_velocity_hz	60.0
pax_measurement_mode	5
pax_wavelength_nm	830.0
phi1_v_lambda	12.2
phi2_v_lambda	12.2
phi1_actuator_volts_per_rp_volt	16.875
phi2_actuator_volts_per_rp_volt	16.875
rp_output_min_voltage	0.0
rp_output_max_voltage	1.0
pd_scope_duration_s	0.01
pd_scope_decimation	64
pd_scope_timeout_s	2.0
stokes_phase_sweep_frequency_hz	0.5
stokes_phase_sweep_amplitude_v	0.36
stokes_phase_sweep_offset_v	0.36
stokes_phase_sweep_sample_period_s	0.05